



NEUTRON & X-RAY OPTICS

Shaping, focusing, and timing the beam

Mirrors, lenses, crystals, multilayers and choppers in McStas & McXtrace

McStas — neutrons

McXtrace — X-rays



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What we'll cover

Everything between source and sample that reshapes the beam in space, direction, energy or time.

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Five ways to shape a beam

A quick map of the optics families we'll cover

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The shared physics behind guides and mirrors

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One interference effect, two very different jobs

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Choppers, velocity selectors, and why neutrons are special



Five ways to shape a beam

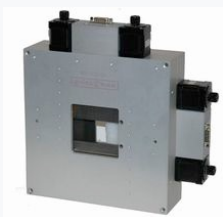
Every optical component in either code falls into one of these families.

Spatial	Reflective	Refractive	Diffractive	Temporal
1 Pinholes, slits, apertures, collimators — define where the beam is allowed to be.	2 Mirrors & guides — steer or focus at grazing incidence.	3 Compound lenses (CRLs) — bend the beam like an ordinary lens, just barely.	4 Crystals & multilayers — select wavelength/energy via interference.	5 Choppers & selectors — shape the beam in time, not space.

We'll move roughly in this order — spatial first, temporal last — flagging where neutrons and X-rays share a component and where they don't.

Spatial & directional shaping

Spatial: pinholes, slits, apertures

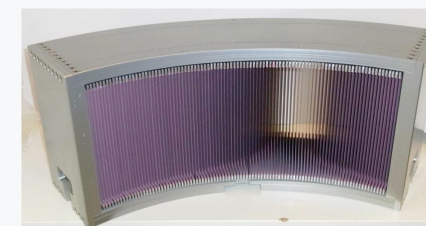


- A static or adjustable “hole” in absorbing material
- Defines the beam's spatial footprint downstream
- Same component idea in both codes: Slit

Directional: collimators



- Two or more slits (or a channel array) define an accepted divergence
- Absorbing walls create parallel “channels” through the collimator
- Usually static, but rotating radial collimators exist — e.g. to reject sample-environment scattering



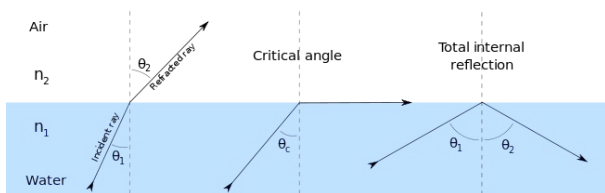
Fully shared: these are geometric, energy-independent devices — the same component names and logic apply whether the beam is neutrons or photons.



Grazing-incidence reflection

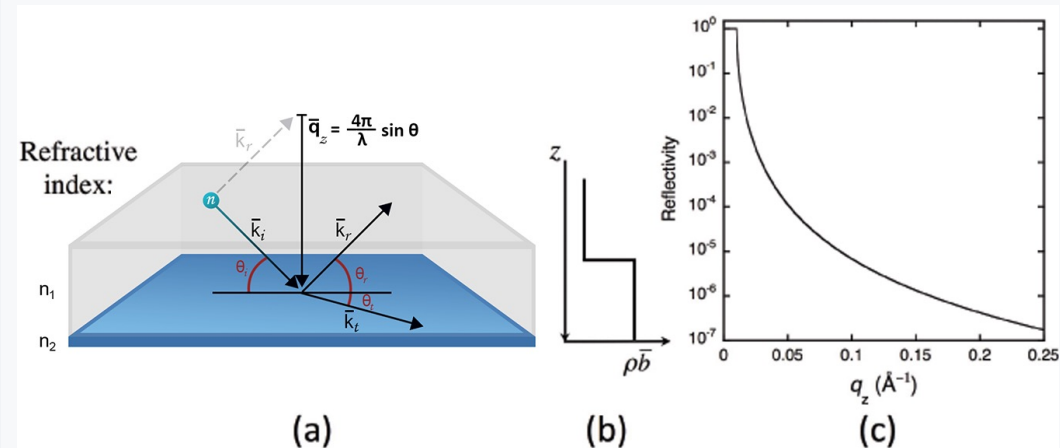
- For both neutrons and X-rays, matter's refractive index is just below 1: $n = 1 - \delta (+ i\beta \text{ for absorption})$.
- Going from vacuum ($n=1$) into a medium with $n<1$ mirrors the classic total-internal-reflection picture — here it's called total external reflection.
- Below a small critical angle θ_c , a ray hitting a smooth, flat surface reflects almost perfectly — no coating chemistry required, just the right angle.
- This single effect underlies both neutron guides and X-ray mirrors — what differs is how it's exploited.

Visible light – water/air interface:
Total **internal** reflection



Neutrons and X-rays – vacuum/matter interface:
Total **external** reflection

Below θ_c : total reflection



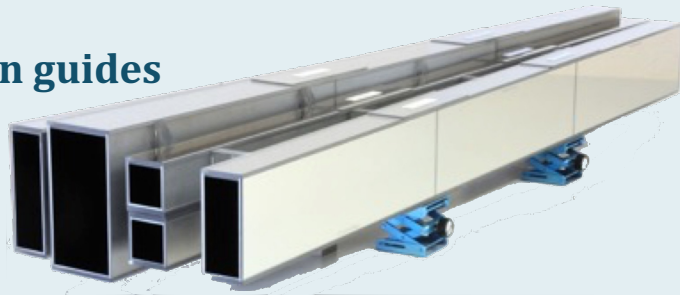
θ_c is tiny — typically a fraction of a degree — for both probes, which is why guides and mirrors are long, shallow, and need to be extremely flat/smooth.



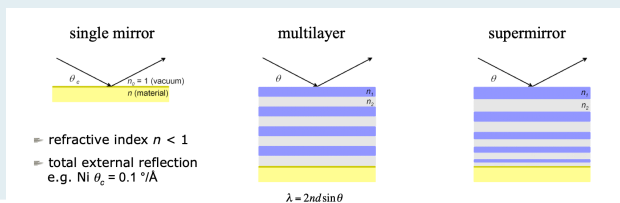
Guides & mirrors

Same critical-angle physics, but neutron beamlines and X-ray beamlines put it to work differently.

McStas · neutron guides

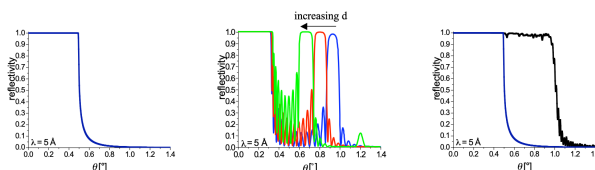


- Guide, Guide_channeled, Guide_gravity: long supermirror-coated channels
- Many shallow bounces over metres — preserves intensity over long transport distances
- Coating quality set by an “m-value” — multiples of the natural-Ni critical angle



McXtrace · X-ray mirrors

- Mirror_curved, TwinKB_ML: one or two large, single-bounce focusing mirrors
- Typically one bounce per axis (e.g. a Kirkpatrick-Baez pair) rather than a many-bounce channel
- Reflectivity data files (measured or computed) describe the coating's angle/energy response



Why the difference? Neutron sources are weak per-neutron, so long guides trade length for preserved flux; X-ray sources are so bright that a couple of well-figured mirrors are enough to focus or steer the beam.

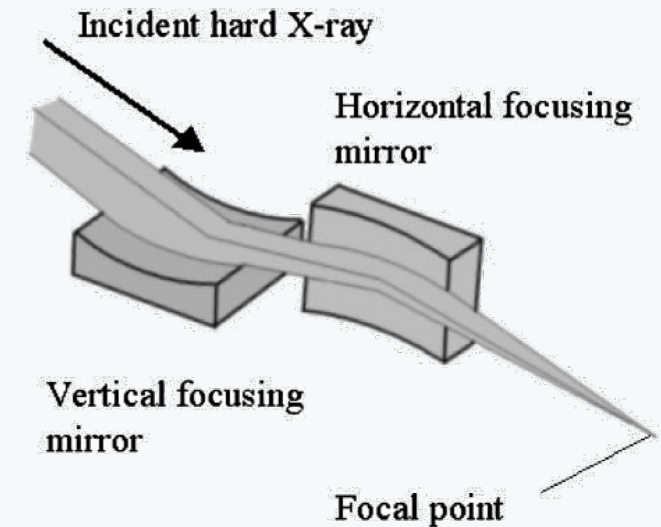
KB mirrors, in practice

A Kirkpatrick-Baez (KB) pair: two perpendicular curved mirrors, one per focusing axis.

- First (meridional) mirror focuses one axis; a glancing angle θ redirects the beam by 2θ .
- An Arm component re-establishes the optical axis after each reflection — it does nothing physically, it just carries a rotated coordinate frame forward.
- The second mirror sits on an Arm rotated 90° around the beam axis, so it focuses the perpendicular direction.
- Focal lengths follow directly from the glancing angle and radius of curvature R:

$$\begin{aligned} f_m &= R \cdot \sin(\theta/2) && \text{(meridional / 1st mirror)} \\ f_s &= R/2 \cdot \sin(\theta) && \text{(sagittal / 2nd mirror)} \end{aligned}$$

source $\rightarrow$ mirror 1 $\rightarrow$ mirror 2 $\rightarrow$ focus



ϑ is typically only a few mrad — chosen from the material's critical angle at the working energy.

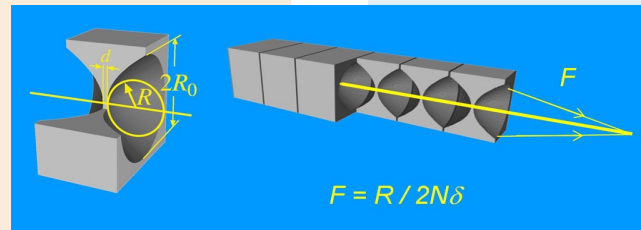
Refractive lenses (CRLs)

Both codes offer compound refractive lenses — and both need them for the same reason.

X-ray

$$\delta_{\text{Xray}} = f(Z, \lambda)$$

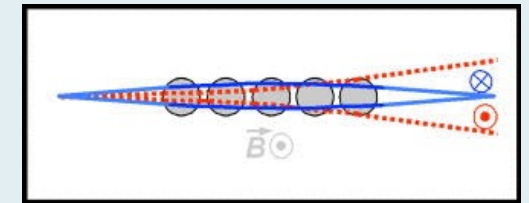
- Grows with electron density — higher atomic number Z means more refraction, but also more absorption
- Low- Z , low-absorption materials preferred: Be, Al, polymers



Neutron

$$n = 1 - \delta_{\text{neutron}} = f(b, B)$$

- Governed by coherent scattering length b , plus a magnetic term B
- The magnetic (spin-dependent) lensing effect is small in practice — hence ordinary CRLs are still the practical choice



Why “compound”: in both regimes the index is barely below 1, so a single lens bends the beam almost imperceptibly. Stacking many (parabolic) lenses in series — a compound refractive lens — adds up to a usable focal length.



A CRL beamline, worked example

Stack	Material	R (tip)	Lenses	Diameter	Thickness (tip)
Stack 1	Be	200 μm	16	1 mm	50 μm
Stack 2	Al	200 μm	10	1 mm	20 μm

Setup: Source_gaussian at E0=23.32 keV (dE=1 keV), CRL stack 31.5 m downstream, focused to a spot 10 m further on. A 2D + energy monitor at the stack checks the incoming footprint and spectrum; another monitor is scanned in z to find the true focus.

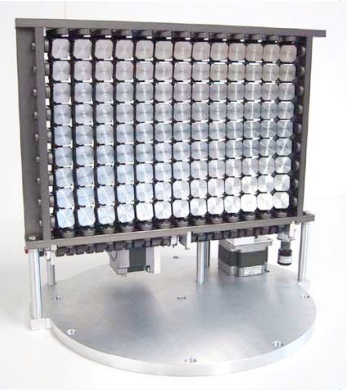
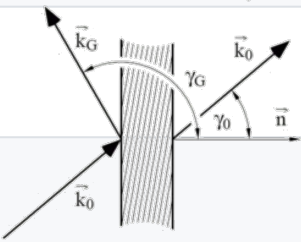
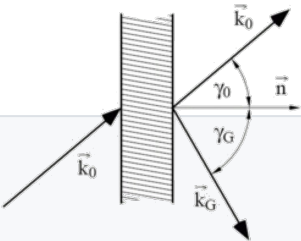
- Directional sampling (focus_xw, focus_yh, dist on the source) concentrates statistics on the aperture that matters, without biasing intensity.
- Used together with a slit, the CRL's achromatic focusing can double as a crude, high-throughput monochromator.
- mxdisplay (“Trace” mode) lets you visually follow rays through the lens stack — useful for sanity-checking alignment.
- Swapping Be ↔ Al lets you compare absorption vs. refraction trade-offs directly.



Crystal monochromators: Bragg's law

`n λ = 2 d sin(θ)` — exactly the same equation, in both codes

	X-ray	Neutron
Typical crystal	Perfect single crystal: Si, Ge, C	Mosaic crystal: pyrolytic graphite, deformed Cu, Ge
Mosaicity	Very low	Often high (multi-crystal, focusing arrays)
Geometry	Laue (transmission) or Bragg (reflection)	Almost always Bragg (reflection)
Resulting beam	Narrow, low divergence and $\frac{\delta\lambda}{\lambda} \approx 0.1\%$	Medium divergence and $\frac{\delta\lambda}{\lambda} \approx 3\%$



In both codes: Bragg_crystal (McXtrace) and Monochromator_flat / Monochromator_curved / Single_crystal (McStas) all rotate to satisfy Bragg's law — the rotation angle is a natural scan parameter for a rocking curve.



Bent crystals & double-crystal monochromators

Bending a crystal

- A radius of curvature on the crystal (Bragg_crystal_bent) is a simple way to model a heat-bump — thermal distortion from absorbed beam power.
- Compare a Divergence_monitor before/after bending to see the extra divergence introduced.
- A rocking curve (scanning θ) shows the peak shape change directly.

Double-crystal monochromators (DCM)

- A second crystal, rotated by the negative of the first angle, brings the beam back parallel to the incoming direction
- Arms track the beam axis between crystals; the pair can share a single rotation stage, as in a real DCM
- Laue_crystal_BC swaps reflection for transmission geometry — useful for higher-energy or higher-heat-load beamlines

```
// crystal 2, mirrored relative to crystal 1's Arm
COMPONENT xtal2 = Bragg_crystal(...)
AT (0,0,0) RELATIVE arm_after_xtal1
ROTATED (-MONO_ANGLE, 0, 0) RELATIVE arm_after_xtal1
```

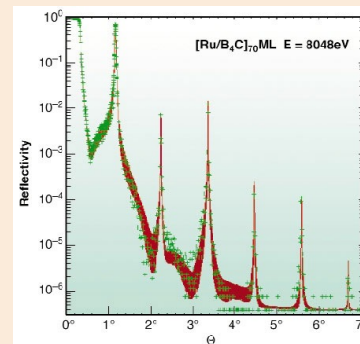
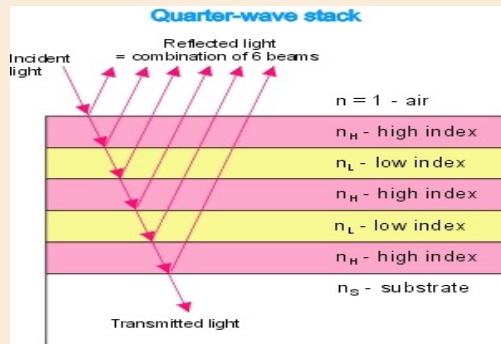


Multilayers: one concept, two jobs

A stack of thin alternating layers reflects by interference — like Bragg's law, but for a synthetic superperiod.

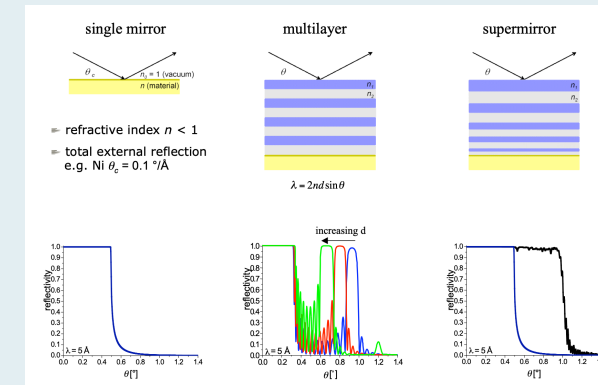
X-ray: a broadband monochromator

- Used as a monochromator with a much broader bandpass — and higher flux — than a perfect crystal
- Pick one of the first “side peaks” on the reflectivity curve
- Curved multilayers combine monochromation with focusing



Neutron: a “supermirror” coating

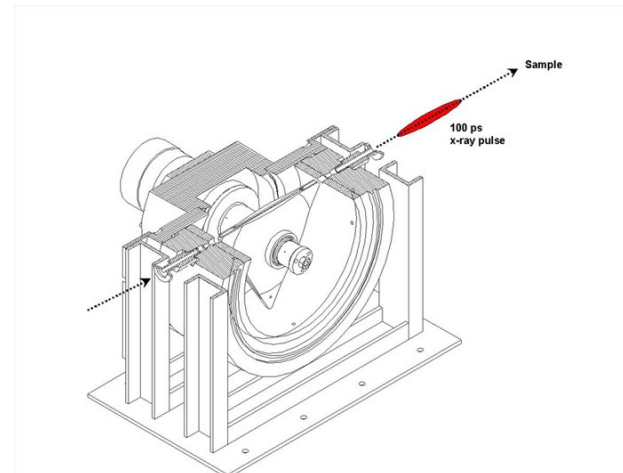
- Stacked Bragg-like reflections extend the total-external-reflection range well beyond a single critical angle
- Ideally a wide, flat reflectivity vs. Q
- Used to coat neutron guides — effectively an “optical fibre” for neutrons



Same underlying phase-interference physics in both codes — only the intended job (monochromate vs. transport) differs.

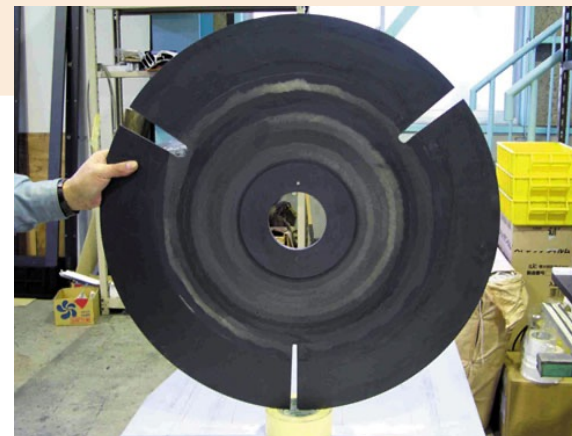
Temporal shaping: choppers

- A rotating absorbing disk with one or more slits cuts a continuous beam into pulses in time.
- Both X-ray and neutron beamlines use a disk-chopper-like component to define or measure time structure — e.g. for time-of-flight (TOF) measurements.
- The physical picture is identical: a slit sweeps across the beam at a set frequency, and only rays arriving within the open window get through.



Where it stops being shared: only neutrons are slow enough that a mechanical rotor can also act as an energy filter. An X-ray chopper shapes time only — it can't pick out an energy band the way a neutron velocity selector or Fermi chopper can.

Up next: two neutron-only rotating devices that filter on speed, not just on time.





Velocity selectors

A rotating drum of twisted, absorbing blades — only a narrow velocity (wavelength) band makes it straight through.



Parameter	Meaning
xwidth, yheight [m]	entry aperture
zdepth [m]	housing length
length, d [m]	blade length / thickness
alpha [deg]	blade twisting angle
radius [m]	rotation axis → aperture-centre distance
nu [Hz]	rotation speed (counterclockwise)
nslit	number of blades

twisted absorbing blades

A schematic diagram showing a series of blades arranged in a circular pattern around a central axis. Arrows indicate the direction of rotation. The blades are twisted, and the diagram illustrates how they filter a beam of light or neutrons.

broad monochromatization
 $\delta\lambda/\lambda \approx 10\%$

A 3D computer-generated image of a velocity selector. It shows a rotating drum with many thin blades. A beam of light or neutrons is shown passing through the blades, with the resulting beam being much narrower and more focused than the incoming beam. The background is dark blue.

MakeAGIF.com

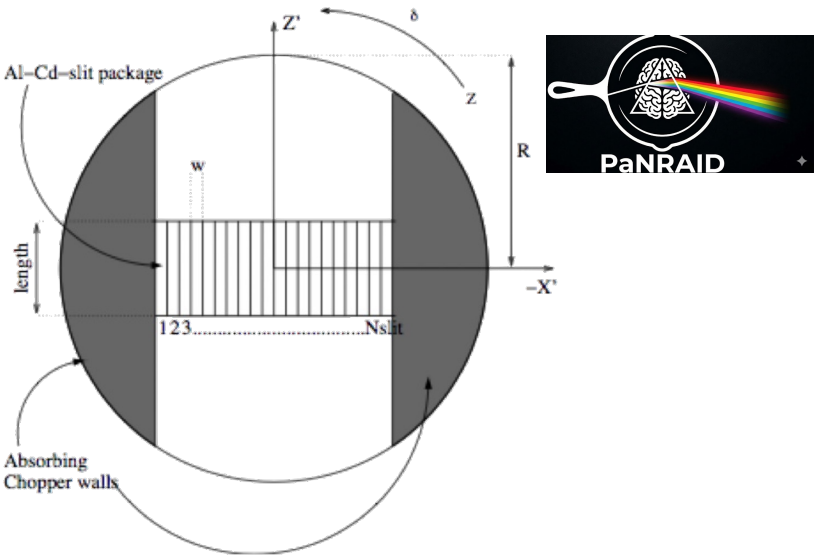
Disk & Fermi choppers

DiskChopper

Parameter	Meaning
<code>nu, theta_0</code>	frequency; angular slit width
<code>radius, yheight, xwidth</code>	disk geometry / slit opening
<code>jitter, delay, phase</code>	timing: random jitter, fixed delay, or angular phase
<code>isfirst, npulse</code>	marks the first chopper; pulses to generate if so

Both are McStas-only: there's no X-ray analogue — the mechanical rotation speeds needed would be absurd at the speed of light. X-ray timing instead comes from the accelerator's bunch structure itself.

Fermi chopper



- A small drum of curved slit packages, spinning fast, sitting directly in the beam — a “rotating collimator”.
- Only neutrons with the right speed can cross the curved channel before it rotates shut — so it acts as a combined fast time-gate and coarse energy filter.
- Used right next to the sample, where a disk chopper's timing alone is no longer selective enough.



One optics toolkit, two beams

What's shared

- Spatial shaping: slits & collimators, same logic either way
- Grazing-incidence total external reflection ($n < 1$, critical angle)
- Bragg's law, and the idea of rotating a crystal to tune it
- Refractive-index-just-below-1 logic behind compound lenses
- Multilayer interference physics
- The chopper concept for defining time structure

What differs

- Many-bounce guides (neutron) vs. one/two-bounce focusing mirrors (X-ray)
- Crystal choice: near-perfect low-mosaicity vs. mosaic/multi-crystal
- Multilayer role: X-ray monochromator vs. neutron guide coating
- Velocity selectors & Fermi choppers exist only for neutrons
- X-ray timing comes from the accelerator, not a mechanical rotor

Where to go deeper

McStas

`mcdoc optics / mcdoc -c`

Guide, Guide_channeled, Monochromator_flat/curved, DiskChopper, FermiChopper, V_selector

mcstas.org

McXtrace

`mxdoc optics / mxdoc -c`

Practical exercises: CRLs, KB mirrors, monochromators — pick two of the three

mcxtrace.org

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